NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

MEGADOOR
ASSA ABLOY

Project: 4 Part, 3 Mullion wire rope hoists

End user: XXXXX

Door Number: XXXX

Method Statement

Installation of a 4 Part Megadoor Hangar Door with wire rope hoist for doors and mullions

DOCUMENT
ISSUE
RECORD

Distribution

Date	Issued to	Holder's Name	Project Role

Revision History

Rev	Written By	Date Checked	Checked by	Doc status	Approved by	Signed	Issue Date

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1. Description of work

1.1. Installation of a 4 Door leaf 3 Mullion VL3190 Megadoor, consisting of 4 wire rope hoists door and 3 wire rope hoist mullions.

2. General

2.1 All employees and self-employed labour will comply with this method statement, with the requirements of the ***Health and Safety at Work Act 1974*** **(UK and Europe)** and all relevant **XXXX local health and safety legislation**, specific site rules and where applicable and the requirements from the principal contractor's safety policy and construction phase health and safety plan will also be adhered to.

2.2 Hold points are specified throughout the process; these are identified by an individual number: HPxx. These numbers correspond to the Hold Point document; in this document the specific requirements to pass the HP must be met and signed off by the appropriate parties.

3. Access to site

3.1 Report to the site office and sign in.

3.2 On arrival at the *XXXXXX* site office all operative will report to Mr (to be confirmed) and will complete a site induction and any other paperwork required.

3.3 A familiarization tour of the site will be conducted.

4. Access to work at height

4.1 Access when working at height will be achieved by using mobile elevating work platforms provided by XXXXXX, all MEWP's supplied and operated according to ***Work at Height Regulations 2005.*** Equipment will be supplied by a reputable national company with all appropriate certification, the HPxx will have been completed on delivery of the plant.

5. Unloading and storage of materials

5.1 Unloading of the door components will be by forklift truck; the operation will be supervised by the Megadoor site supervisor unless alternative arrangements have been made. The fork lift truck will be provided by XXXXXX.

5.2 The components will be positioned within an area designated by the general contractor so as to provide the most economical use of space

and to take into account environmental and safety issues. The positioning of components will be such that the minimum movement from the storage position to the assembly area is taken into account.

5.3 Sensitive components will be stored undercover.

5.4 Using the lifting equipment and the correct manual handling techniques according to ***Manual Handling Operations Regulations 1992 (as amended) and Lifting Operations and Lifting Equipment Regulations 1998.*** Remove the door material from the packaging as appropriate and check all the parts are correct against the packing list.

6 Door Installation

6.1 Installation of components mounted directly to the structure

6.1.1 Complete any permits to work and provide any relevant plant operator certification not already provided.

6.1.2 Familiarise operatives with the working, lay down and restricted areas.

6.1.3 Check that work area is clear from obstructions and tripping/slipping hazards.

6.1.4 Receive and check Forklift and MEWP's including all certificates. *HP1*

6.1.5 Complete a dimensional check of the structural openings and record details. Discuss any deviations with *Steelwork contractor*

6.1.6 A load test of the structure as per Megadoor drawing 65413 Load diagram should been completed by *Steelwork contractor* and the results recorded.

6.1.7 Prepare Hoist unit, Sheave units and mullions for installation

6.1.8 Using the forklift position mullion 12 below its mounting position. Using 20T mobile crane lift the mullion up to its mounting point and fix as per Megadoor installation manual details x & x.

6.1.9 Repeat the operation in 6.1.8 for the installation of mullion 23.

6.1.10 Repeat the operation in 6.1.8 for the installation of mullion 34.

6.1.11 Using the mobile crane lift Hoist unit M01 and place on mounting plate as per Megadoor installation manual detail "attachment plates" (by other than Megadoor) align fixing holes and bolt to the structure, ensure correct torque of the fixing bolts as per installation manual.

6.1.12 Install the mullion hoists for mullion 12 to its mounting position, as shown in installation manual detail “attachment plates”

6.1.13 Install sheave units to Left and right side of the opening for door 1, ensure correct torque of fixing bolts as per installation manual.

6.1.14 Repeat the operation in 6.1.11, 6.1.12 and 6.1.13 for the opening for door 2.

6.1.15 Repeat the operation in 6.1.11, 6.1.12 and 6.1.13 for the opening for door 3.

6.1.16 Repeat the operation in 6.1.11, 6.1.12 and 6.1.13 for the opening for door 4.

6.1.17 Mount Control panel in position show on drawings Position to be confirmed, small adjustments to the position maybe agreed on site. *HP2* to be completed before fixing control panels.

6.1.18 Mains power or a temporary 3 phase supply is required from this point***. Power is by other than AAES Megadoor.***

6.1.19 Fix side guides to fixed jambs

6.1.20 Lay cables in to cable tray. ***Cable trays not by Megadoor***

6.1.21 Cable tie cables to cable tray, leaving ends lose for termination. Run in parallel to 6.1.20 and 6.1.21

6.1.22 Terminate all motor power and brake cables on to field devices as per diagrams 84219001

6.1.23 Terminate all motor and power cables to control panel as per diagram
6.1.24 84219001

6.2 Completion of mullion installation

6.2.1 Fit mullion clip angles and tighten enough to prevent movement of the mullions; carry this out on all 3 mullions.

6.2.2 Install the mullion lifting tubes to mullion 12, 23 and 34. Install and torque up fixing bolts according to installation manual.

6.2.3 Install mullion counter weights to mullion 12, 23 and 34. Install and torque up fixing bolts according to installation manual.

6.2.4 Adjust mullion to mullion pit clearance height and complete the mullion installation by welding the mullion to the clip angles and structure.

6.3 Door Leaf Installation

6.3.1 Using the fork lift position the bottom beam components in door opening 1 assemble the component according to the Megadoor installation manual.

6.3.2 Level the bottom beam in the door opening.

6.3.3 Assemble and stack intermediate beams on top of the levelled bottom beam.

6.3.4 Using the fork lift position the fabric on the beam stack and roll out, line up the fabric on the beam stack centreline.

6.3.5 Commence with the fixing/screwing to the fabric to the top beam.

6.3.6 Using the bottom beam from door 3, position the beam on top of the beam stack and attach the bottom beam to the stack using the attachment brackets provided.

6.3.7 Using the construction program lower the lifting ropes and connect to the fall arrestors of the door 3 bottom beam.

6.3.8 Using the construction program, lift the bottom beam to give the correct pitch between the top beam and next intermediate and screw the fabric to the intermediate beam as before.

6.3.9 Continue with the screwing operation as in 6.3.8 until the fabric has been attached to all the intermediate beams.

6.3.10 Lower the door leaf and remove door 3 bottom beam, attach the lifting ropes to door 1 bottom beam fall arrestors.

6.3.11 Lift the door leaf to the top position using the construction program.

6.3.12 Fix the top intermediate profile to the build using the M12 bolts and lindapters provided. Centralise the door leaf in the opening before torqueing up the fixing bolts.

6.3.13 Repeat tasks 6.3.1- 6.3.12 for door opening 4 use door 1 bottom beam to lift intermediate beams for screwing operation. Once complete re-install door 1 bottom beam.

6.3.14 Position the bottom beams for door 1 and 4 at an appropriate height and complete the fabric screwing operation by fixing the fabric to the bottom beams on door 1 and 3.

6.3.15 Repeat operation 6.3.1-63.12 for door openings 2 and 3 using the using the bottom beams from door 2 and 3 for the screwing operations.

6.3.16 Once the above has been completed repeat operation 6.3.14 for doors 2 and 3.

6.3.17 Fit the bottom seals to all door leafs and trim as required.

6.3.18 Lower the door leafs and mark the stopping position for the wind lock attachment.

6.3.19 Raise the door leafs to allow access to the side guide for the installation of the wind lock lugs.

6.4 Commissioning

6.4.1 Using the construction program lift all doors leafs above the mullion hinge joints.

6.4.2 Check and adjust the mullion lower stop position, complete the installation of the mullion floor plate.

6.4.3 Lift mullion 12 up until it reaches the horizontal position using the construction program, fit the rope stop to ensure it activates the three phase limit switch if the mullion travel above the horizontal position.

6.4.4 Repeat operation 6.4.2 and 6.4.3 on mullion 23.

6.4.5 Repeat operation 6.4.2 and 6.4.3 on mullion 34.

6.4.6 Lower mullion 12 and set the lower limit switch.

6.4.7 Repeat operation 6.4.5 for mullion 23.

6.4.8 Repeat operation 6.4.5 for mullion 34.

6.4.9 Check the mullion lifting ropes have the correct slack when stopped in the fully down position, carry this out on all mullions.

6.4.10 Lower door 1 to the closed position, check it stops correctly on the bottom limit using to normal operation of the controls.

6.4.11 Open door 1 using the normal operation of the controls, when the door reaches the half open position activate the stop button, check if the door stops when the button is activated.

6.4.12 Continue to open door 1 ensure the door leaf stops on the top limit switch.

6.4.13 Repeat operation 6.4.7 – 6.4.10 on door 2.

6.4.14 Repeat operation 6.4.7 – 6.4.10 on door 3.

6.4.15 Repeat operation 6.4.7 – 6.4.10 on door 4.

6.4.16 Open all door fully.

6.4.17 Using normal operation of the control panel test run all mullions from the lower to upper positions; carry out this operation 5 times ensuring that the mullions stop in the horizontal position when reaching the upper stop position, adjust if necessary.

6.5 Testing

6.5.1 Carry out a complete I/O check and check safety interlock functions.

6.5.2 With the mullions in to horizontal position test the door/mullion interlock select each door individually try to close the door by pushing the down button, the doors should not move.

6.5.3 Lower all mullions ensuring they stop on the bottom limit with the correct rope slack, lower door 1 until it clears the top limit switch but not lower than the mullion hinge.

6.5.4 Test the mullion door interlock by selecting mullion 12 and trying to lift the mullion. The mullion should not move.

6.5.5 Completely open door 1 and lower door 2 until it clears the top limit switch but not lower than the mullion hinge.

6.5.6 Test the mullion door interlock by selecting mullion 12 and trying to lift the mullion. The mullion should not move.

6.5.7 Repeat operation 6.5.5 selecting mullion 23.

6.5.8 Completely open door 2 and lower door 3 until it clears the top limit switch but not lower than the mullion hinge.

6.5.9 Repeat operation 6.5.5 selecting mullion 23.

6.5.10 Completely open door 3 and lower door 4 until it clears the top limit switch but not lower then the mullion hinge.

6.5.11 Test the mullion door interlock by selecting mullion 34 and trying to lift the mullion. The mullion should not move.

6.5.12 Lower all mullions and close all doors. Select “all doors” and activate the up button, all doors should open under impulse control and stop in the fully open position.

6.5.13 Select “all mullions” activate the up button the mullions should open under dead mans grip, the movement should stop if the button is

released. Once the mullions reach the horizontal position the controls should stop further upward movement automatically.

6.5.14 Lower all mullions to the fully closed position close all doors to the fully closed position.

6.5.15 Test any external interfaces that are connected; fire alarms or heating interlocks.

6.5.16 Demonstrate and hand over to client carry out operator training and register names of trained operators.

6.5.17 Complete documentation.

6.5.18 Pack away and clear site of all of materials into designated locations.

Appendix A. Supervision, controls and monitoring

1. All work will be directly controlled by the on-site Megadoor Supervisor, who will ensure that the safe means of working as given by this method statement is complied with and that all work platforms are inspected Daily prior to use.

2 The Megadoor Supervisor will liaise with General Contractor concerning any overlapping issues with the General Contractor or other sub-contractors.

Appendix B. Operator training, plant and Tools

Plant

2. Fork lift, 20t crane and 2x Boom lifts

Hand Tools

1. Hammer
2. Screwdriver set
3. Socket Set
4. Combination spanners
5. Spirit Level
6. Laser measure devices
7. Torque wrenches inc calibration certs

Power Tools

1 Angle Grinder
2 Drill
1. Magnetic drill
2. Cordless impacts
3. Welding machine
4. Extension Cable
5. Electric Screwdriver
6. Task Light

7. All operatives will attend and comply with the requirements of any site induction training given by the client or principle contractor.
8. All operatives employed or sub-contracted to this company will be experienced and trained in the work they are undertaking. No employee shall undertake any work that they are not trained to do. Experienced operatives will supervise any trainees. Further training will be provided as required.

9. Plant and machinery will only be operated by those competent and adequately trained in its use on production of the appropriate certification.

10. Power tools and hand tools will be checked before use and at regular intervals for any visual signs of damage. Damaged tools will not be used.

Appendix C. Safety of third parties

1. All employees and sub-contractors will be made aware during site induction of any risk their work can impose of others in the vicinity. Work will only be undertaken so as to minimise these risks or when adequate protection is in place. The site supervisor will liase with the principle contractor to ensure protection against falling materials is adequate.

Appendix D. Personal protective equipment (PPE), first aid and emergency procedures

1 All employees and subcontractors will wear protective footwear, safety helmets and Hi Visibility vest/jacket whilst on site.

2. Other PPE such as gloves, hearing protection, eye protection and respiratory protection and life jackets will be used as required.

3. All operatives will be made aware of the first aid and emergency procedures at the site induction on their arrival. The principle contractor will provide this information.

4. Adequate supplies of fresh clean drinking water are to be provided by the General contractor to ensure hydration and prevent heat stroke to all Megadoor installation teams.

5. Adequate shade and areas for rest are to be provided by the General contractor to ensure the welfare of all Megadoor installation teams.

6. Adequate Welfare/toilet facilities are to be provided for all Megadoor installation teams.

Appendix E. Environmental controls

1. Every effort will be made to keep noise, dust and waste to a minimum level and to ensue that they do not become a hazard or a nuisance to others. Working

areas will be tidied regularly and waste will be removed to skips provided by (other than Megadoor).

Appendix F. Non-standard activities.

1. Should any work become non-standard, either in regard to the materials used or the type of activity to be carried, an appendix will be added to this method statement. This will include necessary risk and COSHH assessments and changes to the method statement. This will ensure the work is carried out safely and in accordance with any site rules.